IC104 Introductory Quantitative Techniques for Business and Finance

Lecture 5. Matrix Algebra

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Lecture Plan

- An Algebraic Definition (Sydsaeter & Hammond Chapter 12.1-12.2)
- Basic Matrix Arithmetic (Sydsaeter & Hammond Chapter 12.2-12.5)
- Matrix Inversion (Sydsaeter & Hammond Chapter 12.6-12.7)
- System of Equations (Sydsaeter & Hammond Chapter 12.6)
- Determinants (Sydsaeter & Hammond Chapter 13.1, 13.2, 13.4, 13.7, 13.8)

Why is Matrix algebra important?

- This section may be long and difficult.
- But - no pain, no gain.
- Matrix algebra forms the heart of the mathematics of econometrics, especially through its application to regressions. You can't understand what's really going on with regression and how the formulae work, until you understand matrix algebra.
- Matrix algebra also forms the core of Asset Management theory - all the main theories of portfolio construction are simply applied matrix algebra. If you take Portfolio Management modules later on, knowing matrix algebra will be very helpful.

System of Equations

- Suppose we have n unknown variables $x_1, x_2, \cdots, x_n$ in m linear equations. We usually write such a system in the form:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots & \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

- $a_{11}, a_{12}, \cdots, a_{mn}$ are called *coefficients* of the system.
- $b_1, b_2, \cdots, b_m$ are called the *right-hand sides*.
- A **solution** of the system is an ordered set or list of numbers $s_1, s_2, \cdots, s_n$ that satisfies all the equations simultaneously when $x_1 = s_1,\ x_2 = s_2,\ \cdots\ , x_n = s_n$.
- If the system has at least one solution, it is said to be **consistent**. When the system has no solution, it is said to be **inconsistent**.

Matrix Definitions

- A **matrix** is a rectangular array of numbers considered as one mathematical object
  - For example: $\mathbf{A}_{(4\times3)} = \begin{pmatrix} 1 & -2 & 3 \\ 4 & -5 & -6 \\ 7 & 8 & 9 \\ 0 & 0 & 10 \end{pmatrix}$
  - This matrix, with 4 rows and 3 columns, is of order 4 by 3. For clarity, we will often show the order (dimension) in parentheses below a matrix.
  - Note that a matrix is represented by a bold, upper case letter.

Matrix Definitions

- A more general example:

$$\mathbf{A}_{(m\times n)} = (a_{ij})_{(m\times n)} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

- This matrix is of order (dimension) $m \times n$.
- Matrix dimension is the number of rows (m) and the number of the columns (n) of the table.
- Elements (entries) in a matrix are numbered by row and column, eg. a_{ij} is the element in row i, column j.
- An individual number, or a 1×1 matrix, is called a scalar.

Special Matrices

- **Square Matrices**: entries a_{ii} are called the *diagonal entries*. The others are called *non-diagonal entries*.

$$\mathbf{A}_{(3\times 3)} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

- A **diagonal matrix** is a square matrix whose non-diagonal elements are zero.

$$\mathbf{A}_{(4\times 4)} = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

Special Matrices

- A square matrix is **lower triangular** if all entries above its main diagonal are 0; for example

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 5 & 2 & 0 \\ 2 & 0 & 3 \end{pmatrix}$$

- Similarly, an **upper-triangular** matrix has zeros below the main diagonal; for example

$$\mathbf{U} = \begin{pmatrix} 5 & 5 & 2 \\ 0 & 2 & -4 \\ 0 & 0 & 3 \end{pmatrix}$$

Special Matrices

- The **identity matrix** of dimension n, denoted $\mathbf{I}_n$, is the $n \times n$ matrix with 1s on the diagonal and 0s elsewhere.

$$\mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- A **row vector** is a $1 \times n$ matrix. Note, lower case letters are used to represent vectors.

$$b_{(1\times n)} = \begin{pmatrix} b_1 & b_2 & \cdots & b_n \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$$

- A **column vector** is an $n \times 1$ matrix.

$$a_{(n\times 1)} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}, \qquad \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

- A **zero matrix** is a matrix consisting only of zeros.

Equality and Product of a Scalar

- Matrix A and B are said to be **equal** if
  - A and B have the same dimensions: $\mathbf{A}_{(m\times n)} = \mathbf{B}_{(m\times n)}$
  - and $a_{ij} = b_{ij}$ for all $i = 1, 2, \cdots, m$ and $j = 1, 2, \cdots, n$
  - When is $\begin{pmatrix} 3 & t-1 \\ 2t & u \end{pmatrix} = \begin{pmatrix} t & 2v \\ u+1 & t+w \end{pmatrix}$?
    - We must have $3 = t$, $t - 1 = 2v$, $2t = u + 1$, and $u = t + w$
    - $t = 3$, $v = 1$, $u = 5$ and $w = 2$
- If α is a real number, we define αA by:

$$\alpha\mathbf{A} = \alpha(a_{ij})_{(m\times n)} = (\alpha a_{ij})_{(m\times n)} = \begin{pmatrix} \alpha a_{11} & \alpha a_{12} & \cdots & \alpha a_{1n} \\ \alpha a_{21} & \alpha a_{22} & \cdots & \alpha a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha a_{m1} & \alpha a_{m2} & \cdots & \alpha a_{mn} \end{pmatrix}$$

Matrix Addition and Subtraction

- If $\mathbf{A} = (a_{ij})_{m\times n}$ and $\mathbf{B} = (b_{ij})_{m\times n}$ are two matrices of the same order, we define the sum of A and B as the $m \times n$ matrix $(a_{ij} + b_{ij})_{m\times n}$:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \\ a_{31} + b_{31} & a_{32} + b_{32} & a_{33} + b_{33} \end{pmatrix}$$

- We simply add each respective element of the two matrices.

Matrix Addition and Subtraction

- Rules for matrix addition and multiplication by scalars:

- $(\mathbf{A}+\mathbf{B})+\mathbf{C} = \mathbf{A}+(\mathbf{B}+\mathbf{C})$
- $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
- $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B}) = -(\mathbf{B} - \mathbf{A})$
- $\mathbf{A} + (-\mathbf{A}) = \mathbf{0}$
- $\mathbf{A} + \mathbf{0} = \mathbf{A}$
- $-(-\mathbf{A}) = \mathbf{A}$
- $(\alpha + \beta)\mathbf{A} = \alpha\mathbf{A} + \beta\mathbf{A}$
- $\alpha(\mathbf{A} + \mathbf{B}) = \alpha\mathbf{A} + \alpha\mathbf{B}$

Example

- Compute $\mathbf{A}+\mathbf{B}$, $3\mathbf{A}$, and $(-\frac{1}{2})\mathbf{B}$, if $\mathbf{A} = \left(\begin{smallmatrix} 1 & 2 & 0 \\ 4 & -3 & -1 \end{smallmatrix}\right)$ and $\mathbf{B} = \left(\begin{smallmatrix} 0 & 1 & 2 \\ 1 & 0 & 2 \end{smallmatrix}\right)$

$$\mathbf{A}+\mathbf{B} = \begin{pmatrix} 1+0 & 2+1 & 0+2 \\ 4+1 & -3+0 & -1+2 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 2 \\ 5 & -3 & 1 \end{pmatrix}$$

$$3\mathbf{A} = \begin{pmatrix} 3\cdot 1 & 3\cdot 2 & 3\cdot 0 \\ 3\cdot 4 & 3\cdot(-3) & 3\cdot(-1) \end{pmatrix} = \begin{pmatrix} 3 & 6 & 0 \\ 12 & -9 & -3 \end{pmatrix}$$

$$(-\frac{1}{2})\mathbf{B} = \begin{pmatrix} -\frac{1}{2}\cdot 0 & -\frac{1}{2}\cdot 1 & -\frac{1}{2}\cdot 2 \\ -\frac{1}{2}\cdot 1 & -\frac{1}{2}\cdot 0 & -\frac{1}{2}\cdot 2 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{2} & -1 \\ -\frac{1}{2} & 0 & -1 \end{pmatrix}$$

Matrix Multiplication

- Let $\mathbf{A} = (a_{ij})_{m \times \mathbf{n}}$ and $\mathbf{B} = (b_{ij})_{\mathbf{n} \times p}$. Then their product $\mathbf{C} = \mathbf{AB}$ is an $m \times p$ matrix such that

$$c_{ij} = \sum_{r=1}^{n} a_{ir} b_{rj} = a_{i1} b_{1j} + a_{i2} b_{2j} + \cdots + a_{ik} b_{kj} + \cdots + a_{in} b_{nj}$$

- We take a row in the first matrix, and multiply each element of it by the elements in a column of the second matrix, to produce one element of the output matrix.
- The location of this resulting element in the output matrix has the row number of the row that we've just used from the first matrix, and the column number of the column that we've just used from the second matrix.

Matrix Multiplication

$$\begin{pmatrix} a_{11} & \cdots & a_{1k} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ik} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mk} & \cdots & a_{mn} \end{pmatrix} \cdot \begin{pmatrix} b_{11} & \cdots & b_{1j} & \cdots & b_{1p} \\ \vdots & & \vdots & & \vdots \\ b_{k1} & \cdots & b_{kj} & \cdots & b_{kp} \\ \vdots & & \vdots & & \vdots \\ b_{n1} & \cdots & b_{nj} & \cdots & b_{np} \end{pmatrix} =$$

$$\begin{pmatrix} c_{11} & \cdots & c_{1j} & \cdots & c_{1p} \\ \vdots & & \vdots & & \vdots \\ c_{i1} & \cdots & c_{ij} & \cdots & c_{ip} \\ \vdots & & \vdots & & \vdots \\ c_{m1} & \cdots & c_{mj} & \cdots & c_{mp} \end{pmatrix}$$

Example I

- $\mathbf{A} = \begin{pmatrix} 0 & 1 & 2 \\ 2 & 3 & 1 \\ 4 & -1 & 6 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 3 & 2 \\ 1 & 0 \\ -1 & 1 \end{pmatrix}$

$$\Rightarrow \mathbf{AB} = \begin{pmatrix} 0 & 1 & 2 \\ 2 & 3 & 1 \\ 4 & -1 & 6 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 8 & 5 \\ 5 & 14 \end{pmatrix}$$

- Is product **BA** defined?

Example II

$$\mathbf{A} = \begin{pmatrix} 1 & -5 & 3 \\ 0 & -2 & 6 \\ 7 & 2 & -4 \end{pmatrix} \mathbf{B} = \begin{pmatrix} -8 & 6 & 1 \\ 7 & 0 & -3 \\ 2 & 4 & 5 \end{pmatrix}$$

$$\mathbf{AB} = \begin{pmatrix} 1 \cdot (-8) + (-5) \cdot 7 + 3 \cdot 2 & 1 \cdot 6 + (-5) \cdot 0 + 3 \cdot 4 & 1 \cdot 1 + (-5) \cdot (-3) + 3 \cdot 5 \\ 0 \cdot (-8) + (-2) \cdot 7 + 6 \cdot 2 & 0 \cdot 6 + (-2) \cdot 0 + 6 \cdot 4 & 0 \cdot 1 + (-2) \cdot (-3) + 6 \cdot 5 \\ 7 \cdot (-8) + 2 \cdot 7 + (-4) \cdot 2 & 7 \cdot 6 + 2 \cdot 0 + (-4) \cdot 4 & 7 \cdot 1 + 2 \cdot (-3) + (-4) \cdot 5 \end{pmatrix}$$

$$\Rightarrow \mathbf{AB} = \begin{pmatrix} -37 & 18 & 31 \\ -2 & 24 & 36 \\ -50 & 26 & -19 \end{pmatrix}$$

Example III

$$\underset{(1\times 4)}{\begin{pmatrix}\beta_0 & \beta_1 & \beta_2 & \beta_3\end{pmatrix}} \cdot \underset{(4\times 1)}{\begin{pmatrix}1\\ x_1\\ x_2\\ x_3\end{pmatrix}} = \underset{(1\times 1)}{\begin{pmatrix}\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3\end{pmatrix}}$$

$$\underset{(4\times 1)}{\begin{pmatrix}1\\ x_1\\ x_2\\ x_3\end{pmatrix}} \cdot \underset{(1\times 4)}{\begin{pmatrix}\beta_0 & \beta_1 & \beta_2 & \beta_3\end{pmatrix}} = \underset{(4\times 4)}{\begin{pmatrix}\beta_0 & \beta_1 & \beta_2 & \beta_3\\ x_1\beta_0 & x_1\beta_1 & x_1\beta_2 & x_1\beta_3\\ x_2\beta_0 & x_2\beta_1 & x_2\beta_2 & x_2\beta_3\\ x_3\beta_0 & x_3\beta_1 & x_3\beta_2 & x_3\beta_3\end{pmatrix}}$$

The Transpose

- Consider a matrix $\mathbf{A}_{m\times n}$. The transpose of $\mathbf{A}$, denoted by $\mathbf{A}'$ or $\mathbf{A}^T$, is defined as the $n \times m$ matrix whose first column is the first row of $\mathbf{A}$, whose second column is the second row of $\mathbf{A}$, and so on. For example:

$$\mathbf{X}_{(4\times 3)} = \begin{pmatrix} 1 & -2 & 3 \\ 4 & -5 & 6 \\ 7 & 8 & 9 \\ 0 & 0 & 10 \end{pmatrix}, \quad \mathbf{X}'_{(3\times 4)} = \begin{pmatrix} 1 & 4 & 7 & 0 \\ -2 & -5 & 8 & 0 \\ 3 & -6 & 9 & 10 \end{pmatrix}$$

- In general:

$$\mathbf{A}_{(m\times n)} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{A}'_{(n\times m)} = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}$$

Symmetric Matrix

- Square matrices with the property that they are symmetric about the main diagonal are called **symmetric**. For example:

$$\begin{pmatrix} -3 & 2 \\ 2 & 0 \end{pmatrix}, \quad \begin{pmatrix} 2 & -1 & 5 \\ -1 & -3 & 2 \\ 5 & 2 & 8 \end{pmatrix}, \quad \begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix}$$

- The matrix $\mathbf{A}$ is symmetric $\Leftrightarrow \mathbf{A} = \mathbf{A}'$

Properties of Matrix Multiplication and Transpose

- $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$ (associative law)
- $\mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB}+\mathbf{AC}$ (left distributive law)
- $(\mathbf{A}+\mathbf{B})\mathbf{C} = \mathbf{AC}+\mathbf{BC}$ (right distributive law)
- $(\alpha\mathbf{A})\mathbf{B} = \mathbf{A}(\alpha)\mathbf{B} = \alpha(\mathbf{AB})$
- $\mathbf{A}_{m\times n}\mathbf{I}_n = \mathbf{I}_m\mathbf{A}_{m\times n} = \mathbf{A}_{m\times n}$ (multiplication by the Identity matrix leaves it unchanged)
- $\mathbf{A}_{m\times n}\mathbf{0}_{n\times p} = \mathbf{0}_{m\times p}$ (multiplication by the Zero Matrix yields the Zero matrix)
- $(\mathbf{A}')' = \mathbf{A}$
- $(\mathbf{A}+\mathbf{B})' = \mathbf{A}'+\mathbf{B}'$
- $(\alpha\mathbf{A})' = \alpha\mathbf{A}'$
- $(\mathbf{A}_{m\times n}\mathbf{B}_{n\times p})' = \mathbf{B}'_{p\times n}\mathbf{A}'_{n\times m}$ (transpose of a product)
- $(\mathbf{AB}\cdots\mathbf{F})' = \mathbf{F}'\cdots\mathbf{B}'\mathbf{A}'$

Notes

- Matrix multiplication is **not** in general commutative, i.e. it's not the same forwards as backwards:
  - For $\mathbf{A}_{m\times n}\mathbf{B}_{n\times p}$, the product $\mathbf{BA}$ is not defined unless $m = p$.
  - For $\mathbf{A}_{m\times n}$ and $\mathbf{B}_{n\times m}$, the product $\mathbf{AB}$ is $(m \times m)$ and $\mathbf{BA}$ is $(n \times n)$, so they can not be equal unless $m = n$.
  - Even when $\mathbf{A}$ and $\mathbf{B}$ are both $(n \times n)$, the products $\mathbf{AB}$ and $\mathbf{BA}$ need not be equal.
  - When $\mathbf{AB}=\mathbf{BA}$, the matrices $\mathbf{A}$ and $\mathbf{B}$ are said to commute.

System of Equations in Matrix Form

- The definition of matrix multiplication makes it simple to formulate systems of scalar equations as a single matrix equation, often providing a useful level of abstraction.
- For example, consider the following system of two linear equations in two unknowns, x_1 and x_2:

$$\begin{aligned} 2x_1 + 5x_2 &= 4 \\ x_1 + 3x_2 &= 5 \end{aligned}$$

- We can write these equations as a matrix equation,

$$\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$

$$\mathbf{A}_{(2\times 2)}\mathbf{X}_{(2\times 1)} = \mathbf{b}_{(2\times 1)}$$

System of Equations in Matrix Form

- More generally, for m linear equations in n unknowns,

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\
\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots & \\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m
\end{aligned}
$$

- The system of equations can be written as

$$
\mathbf{A}_{(m\times n)}\mathbf{x}_{(n\times 1)} = \mathbf{b}_{(\mathrm{m}\times 1)}
$$

$$
\mathbf{A}_{(m\times n)} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{x}_{(n\times 1)} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b}_{(\mathrm{m}\times 1)} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}
$$

Matrix Inversion

- In scalar algebra, division is essential to the solution of linear equations. We rely on dividing both the Left Hand Side and the Right Hand Side of an equation by the same number to solve it. For example:

$$6x = 12 \Rightarrow x = \frac{12}{6} = 2$$

$$\text{Or } \frac{1}{6} \cdot 6x = x = \frac{1}{6} \cdot 12 = 2$$

$$\text{where } \frac{1}{6} = 6^{-1} \text{ is the scalar inverse of 6}$$

Matrix Inversion

- In matrix algebra, there is no direct analog of division, but most square matrices have a matrix inverse.
- Given a matrix $\mathbf{A}_{(n\times n)}$, $\mathbf{X}_{(n\times n)}$ is an inverse of $\mathbf{A}$ if

$$\mathbf{A}_{(n\times n)}\mathbf{X}_{(n\times n)} = \mathbf{X}_{(n\times n)}\mathbf{A}_{(n\times n)} = \mathbf{I}_{(n\times n)}$$

- We can write the inverse of $\mathbf{A}_{(n\times n)}$ as $\mathbf{A}^{-1}_{(n\times n)}$.
- We can think of the inverse of a matrix as that which, when multiplied together with it, will produce the Identity matrix.

Matrix Inversion

- Only **square** matrices can have inverses. But not even all square matrices have inverses.
- Square matrices that have an inverse are termed **nonsingular**.
- Squares matrices with no inverse are **singular** (i.e., strange, unusual).
- If an inverse of a square matrix exists, it is unique.
- Moreover, if, for square matrices $\mathbf{A}$ and $\mathbf{B}$, $\mathbf{AB} = \mathbf{I}$, then necessarily $\mathbf{BA} = \mathbf{I}$, and $\mathbf{B} = \mathbf{A}^{-1}$.
- In scalar algebra, only the number 0 has no inverse (i.e., $\mathbf{0}^{-1} = \frac{1}{0}$ is undefined). In matrix algebra, $0_{n\times n}$ is singular, but there are also nonzero singular matrices.

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

$$\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} \\ 0 & 0 \end{pmatrix} \neq \mathbf{I}_2$$

Matrix Inversion Properties

- A useful fact to remember is that if the matrix $\mathbf{A}$ is singular then at least one column of $\mathbf{A}$ can be written as a multiple of another column or as a weighted sum (linear combination) of the other columns; and, equivalently, at least one row can be written as a multiple of another row or as a weighted sum of the other rows.
  - In the example above, the second row is 0 times the first row, and the second column is 0 times the first column.
  - Indeed, any square matrix with a zero row or column is therefore singular.
- Let $\mathbf{A}$ and $\mathbf{B}$ be invertible $n \times n$ matrices. Then:
  - $\mathbf{A}^{-1}$ is invertible, and $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$
  - $\mathbf{AB}$ is invertible, and $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
  - $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$
  - $(c\mathbf{A})^{-1} = c^{-1}\mathbf{A}^{-1}$, where $c \neq 0$

Application of Matrix Inversion

- The inverse of the nonsingular matrix $\mathbf{B} = \left(\begin{smallmatrix} 2 & 5 \\ 1 & 3 \end{smallmatrix}\right)$ is the matrix $\mathbf{B}^{-1} = \left(\begin{smallmatrix} 3 & -5 \\ -1 & 2 \end{smallmatrix}\right)$

$$\text{Check: } \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

- Matrix inverses are useful for solving systems of linear simultaneous equations where there is an equal number of equations and unknowns.
  - The general form of such a problem, with n equations and n unknowns, is $\mathbf{A}_{(n\times n)}\mathbf{X}_{(n\times 1)} = \mathbf{b}_{(n\times 1)}$
  - The solution is obtained by multiplying both sides of the equation on the left by the inverse of the coefficient matrix:

$$\mathbf{A}^{-1}\mathbf{A}\mathbf{X} = \mathbf{A}^{-1}\mathbf{b} \Rightarrow \mathbf{I}\mathbf{X} = \mathbf{A}^{-1}\mathbf{b} \Rightarrow \mathbf{X} = \mathbf{A}^{-1}\mathbf{b}$$

Example

- Consider the following system of 2 equations in 2 unknowns:

$$\begin{aligned} 2x_1 + 5x_2 &= 4 \\ x_1 + 3x_2 &= 5 \end{aligned}$$

- In matrix form:

$$\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}^{-1} \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ 5 \end{pmatrix} = \begin{pmatrix} -13 \\ 6 \end{pmatrix}$$

- That is, the solution is $x_1 = -13$, $x_2 = 6$.

Determinants

- The determinant of a square matrix is a scalar that can be computed from the elements of the square matrix.
- The determinant of a matrix $\mathbf{A}$ is denoted by $\det(\mathbf{A})$, $\det \mathbf{A}$ or $|\mathbf{A}|$.
- The determinant helps us find the inverse of a matrix and is useful in calculus as well as in solving systems of linear equations.

Determinant Formula

- For a general 2×2 matrix $\mathbf{A}$, its determinant is defined as

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad |\mathbf{A}| = a_{11}a_{22} - a_{21}a_{12}$$

- For example:

$$\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}, \qquad |\mathbf{A}| = 4 \cdot 2 - 3 \cdot 1 = 5$$

- For a general 3×3 matrix $\mathbf{A}$, its determinant is defined as:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\begin{aligned} |\mathbf{A}| = {} & a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} + a_{12}a_{23}a_{31} \\ & - a_{12}a_{21}a_{33} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} \end{aligned}$$

Determinant and Cofactor

- We can write the determinant formula for a 3×3 matrix as:

$$
\begin{aligned}
|\mathbf{A}| &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \\
&= a_{11} \cdot \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \cdot \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \cdot \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}
\end{aligned}
$$

Determinant and Cofactor

- Each element in the first row is multiplied by its cofactors in the parentheses following it. Note that each cofactor is (plus or minus) the determinant of a two by two matrix. That determinant is made up of products of elements in the rows and columns NOT containing a_{1j}.

$$\begin{pmatrix} a_{11} & . & . \\ . & a_{22} & a_{23} \\ . & a_{32} & a_{33} \end{pmatrix}$$

$$\begin{pmatrix} . & a_{12} & . \\ a_{21} & . & a_{23} \\ a_{31} & . & a_{33} \end{pmatrix}$$

$$\begin{pmatrix} . & . & a_{13} \\ a_{21} & a_{22} & . \\ a_{31} & a_{32} & . \end{pmatrix}$$

Determinant and Cofactor

- In general, we define the cofactor of a_{ij} as C_{ij}. C_{ij} equals $(-1)^{i+j}$ times the determinant of the $n-1$ by $n-1$ square matrix obtained by removing row i and column j.
- C_{ij} is positive if $i+j$ is even and negative if $i+j$ is odd.
- For $n \times n$ matrices, the cofactor formula to calculate determinant is:

$$det\mathbf{A} = |\mathbf{A}| = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}$$

- Applying this to a 2×2 matrix gives us:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad |\mathbf{A}| = a_{11}a_{22} + a_{12}a_{21}(-1)$$

Basic Rules for Determinants

- Let $\mathbf{A}$ be an $n \times n$ matrix. Then:
  - If all the elements in a row (or column) of $\mathbf{A}$ are 0, then $|\mathbf{A}| = 0$.
  - $|\mathbf{A}'| = |\mathbf{A}|$
  - If all the elements in a single row (or column) of $\mathbf{A}$ are multiplied by a number α, the determinant is multiplied by α
  - If two rows (or two columns) of $\mathbf{A}$ are interchanged, the determinant changes sign, but the absolute value remains unchanged.
  - If two of the rows (or columns) of $\mathbf{A}$ are proportional, then $|\mathbf{A}| = 0$
  - The value of the determinant of $\mathbf{A}$ is unchanged if a multiple of one row (or one column) is added to a different row (or column) of $\mathbf{A}$.
  - $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$
  - If α is a real number, $|\alpha\mathbf{A}| = \alpha^n|\mathbf{A}|$
  - In general: $|\mathbf{A} + \mathbf{B}| \neq |\mathbf{A}| + |\mathbf{B}|$

A General Formula for the Inverse

- Given the definition of the cofactor, we define the cofactor matrix of $\mathbf{A}$ as C^+ and **adjoint** matrix of $\mathbf{A}$ as $adj(\mathbf{A})$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \quad C^+ = \begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{pmatrix}$$

$$adj(\mathbf{A}) = (C^+)' = \begin{pmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{pmatrix}$$

- Any square matrix $\mathbf{A} = (a_{ij})_{n \times n}$ with determinant $|\mathbf{A}| \neq 0$ has a unique inverse $\mathbf{A}^{-1}$ satisfying $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$. This is given by:

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \cdot adj(\mathbf{A})$$

Example

- Find the inverse for matrix $\mathbf{G}$:

$$\mathbf{G} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

- The inverse of $\mathbf{G}$ is

$$\mathbf{G}^{-1} = \frac{1}{|\mathbf{G}|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{(ad - bc)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Example

- Find the inverse for matrix **A**:

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{pmatrix}$$

- The nine cofactors are:

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 3 & 1 \\ 2 & 4 \end{vmatrix} = 10, \quad C_{12} = (-1)^{1+2} \begin{vmatrix} 4 & 1 \\ 1 & 4 \end{vmatrix} = -15, \quad C_{13} = (-1)^{1+3} \begin{vmatrix} 4 & 3 \\ 1 & 2 \end{vmatrix} = 5$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 3 & 4 \\ 2 & 4 \end{vmatrix} = -4, \quad C_{22} = (-1)^{2+2} \begin{vmatrix} 2 & 4 \\ 1 & 4 \end{vmatrix} = 4, \quad C_{23} = (-1)^{2+3} \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = -1$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 3 & 4 \\ 3 & 1 \end{vmatrix} = -9, \quad C_{32} = (-1)^{3+2} \begin{vmatrix} 2 & 4 \\ 4 & 1 \end{vmatrix} = 14, \quad C_{33} = (-1)^{3+3} \begin{vmatrix} 2 & 3 \\ 4 & 3 \end{vmatrix} = -6$$

- Hence, the inverse of **A** is

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix} = -\frac{1}{5} \begin{pmatrix} 10 & -4 & -9 \\ -15 & 4 & 14 \\ 5 & -1 & -6 \end{pmatrix}$$